Solution Design Document

1. Business Problem

This project studies whether the Piotroski F-Score can be used to identify stronger stocks within high book-to-market equity universes in the United States, Japan, and Vietnam. We first use the F-Score as a stock selection method, and then apply modern portfolio theory to construct portfolios from the selected stocks to see whether portfolio optimization can add further value. Finally, we compare the resulting portfolios against matched random portfolios through Monte Carlo simulation (the so-called *null portfolios*) to assess whether the observed performance is meaningful or could simply be explained by luck.

2. Project Architecture

We built the project based on the following methodological steps.

Step 1. Data collection and processing. Data is gathered from reliable sources, including financial statements, stock prices, and historical membership data where available. The data is then cleaned and missing values, erroneous entries, delistings, and survivorship bias are handled consistently.

Step 2. Derived data and screening. F-Score signals and book-to-market ratios are computed, and the top 40% by B/M is retained.

Step 3. Selection and construction. The top K=20 stocks by F-Score are selected and equal-weight, GMV, and sector-capped GMV portfolios are constructed. Repeated for K=25 and K=30.

Step 4. Price matching and backtest. Prices are matched based on the point-in-time principle, and portfolios are formed on July 1 and held through June 30 the following year.

Step 5. Robustness and benchmarks. The strategies are compared against matched random portfolios through Monte Carlo simulation using a 5% significance level. These are the so-called *null portfolios*.

3. Coding Architecture: Roles, Stage-by-stage

There are five main stages. The table names what each stage does and where its code lives on GitHub.

Stage	Code location	What it does
1. Data	`src/fscore/data/`	Maps U.S., Japan, and Vietnam inputs into a common point-in-time schema. Vietnam enters through `src/fscore`_vietnam/ after local preprocessing.
2. Scoring	`src/fscore/signal/`	Computes B/M and the nine F-Score tests. A firm-year missing a required signal is excluded instead of being treated as a low score.
3. Selection	`src/fscore/selection/`	Keeps the top 40% by B/M, selects K=20 by F-Score with seeded tie-breaking, and draws matched random baskets from the same annual pool. Repeated for K=25 and K=30.
4. Weighting	`src/fscore/construction/`	Constructs equal weight, long-only RMT-denoised GMV, and 20% sector-capped GMV. GMV uses 36 months of daily returns for the covariance matrix; detoning is not used.
5. Evaluation	`src/fscore/evaluation/`	Runs the annual July-to-June backtest and reports return, volatility, Sharpe, maximum drawdown, turnover and Effective N. Performance is gross of costs.
Main pipeline	`src/fscore/pipeline.py` `grid.py`	Runs the five stages end to end for the headline study and supporting sensitivity cases across portfolio sizes and markets.
Reproducibility safeguards	`tests/ (22 tests)` `scripts/`	Tests pin key design choices, while scripts regenerate analysis outputs from the prepared country inputs.

Vietnam data path. FireAnt, CafeF and TCBS data are processed separately in `src/fscore_vietnam/`, including accounting checks and a Vietnam-specific liquidity gate. Shared selection, construction, and evaluation logic as the U.S. and Japan.

4. Key Design Decisions

- **Point-in-time discipline.** All portfolios form annually from 2012 through 2024 using only information available before the July formation date.
- **Value first, F-Score second.** The top 40% by B/M is retained before F-Score ranking; incomplete nine-signal scores are excluded.
- **Long-only construction.** The study uses equal weight plus long-only GMV and long-only sector-capped GMV. RMT denoising is applied to GMV covariance estimates; detoning is not used.
- **Gross performance and matched inference.** Returns are reported gross of transaction costs; turnover is reported separately. Monte Carlo nulls use the same annual high-B/M pool and a 5% significance level.

5. Deliverables and Cost

The project comes with the GitHub repository, notebooks, scripts, tests, README, and the requirements file; derived data panels as well as figures and tables used in the report are also included. No special paid compute infrastructure is required to rerun the shared analysis from prepared inputs. Japan uses Bloomberg-sourced financial statements and Vietnam uses self-built local data, hence raw licensed or source data are not redistributed.